

MATH 220/199 Week 12 Day 2

Reminders

1. Office hours are held in Davenport 330 at 3pm on Thursdays.
2. Do the Merit Quiz on the Merit Canvas page.

One of these Guys is a Trig Guy...

INSTRUCTIONS: Today's worksheet should give you the heuristics/intuition to pick out whether a derivative has a trigonometric anti-derivative (or not). The problems come in pairs, one will be a trig anti-derivative, the other will not be. Compute both. (HINT: You can verify your answer by taking the derivative).

1. Pair 1:

(a) $f'(x) = \frac{2x}{\sqrt{1-x^4}}$

(b) $g'(x) = -\frac{2x}{(1+x^2)^2}$

2. Pair 2:

(a) $f'(x) = \frac{1}{(1+x^2)+1}$

(b) $g'(x) = -\frac{2}{(2x-1)^2}$

3. Pair 3 (HARD):

(a) $f'(x) = \frac{1}{2(\sqrt{x}+1)^2\sqrt{x}}$

(b) $g'(x) = -\frac{1}{2\sqrt{-x^2+x}}$

Sketching

Each of the graphs below represents some function $f'(x)$. To the right of each graph, make a sketch of a possible graph for $f(x)$.

